

Qwen3.5-9B UK: military status — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	21.11 [17.11, 25.11]				5.37	14.89	0.11	6.09	30.62	79.33	0.60	4.22
SFT (LoRA, merged)	implicit	2.22 [0.89, 3.56] ♦	-18.89 ♦	89 : 4	6e-22	0.37 (5.37)	1.11 (14.89)	0.01 (0.11)	0.53 (6.09) ♦	12.09 (30.62)	75.29 (79.33)	0.65 (0.60)	0.00 (4.22) ♦
LEACE (layer 16)	implicit	18.36 [13.26, 23.64] (own baseline 27.34, n=256)	-8.98	55 : 32	0.02	4.50 (5.37)	15.68 (14.89)	0.12 (0.11)	5.39 (6.09)	45.82 (30.62)	72.45 (79.33)	0.59 (0.60)	1.87 (4.22)
scrub: llm	implicit	12.44 [9.33, 15.78]	-8.67	60 : 21	2e-05	0.36 (5.37) ♦	0.89 (14.89) ♦	0.01 (0.11) ♦	3.82 (6.09)	31.78 (30.62)	82.18 (79.33) ♦	0.62 (0.60)	0.09 (4.22)
prompt: structured_rubric	implicit	14.67 [11.11, 18.22]	-6.44	59 : 30	3e-03	3.24 (5.37)	10.89 (14.89)	0.07 (0.11)	3.78 (6.09)	20.36 (30.62)	77.42 (79.33)	0.55 (0.60)	1.87 (4.22)
prompt: recruiter_guidelines	implicit	14.89 [11.33, 18.44]	-6.22	45 : 17	5e-04	3.11 (5.37)	10.00 (14.89)	0.07 (0.11)	4.13 (6.09)	25.78 (30.62)	79.91 (79.33)	0.62 (0.60)	3.73 (4.22)
prompt: counterfactual_invariance	implicit	15.11 [11.33, 18.67]	-6.00	36 : 9	7e-05	1.80 (5.37)	5.11 (14.89)	0.05 (0.11)	4.27 (6.09)	31.42 (30.62)	80.22 (79.33)	0.61 (0.60)	2.44 (4.22)
prompt: fairness_constitution	implicit	15.56 [12.22, 19.33]	-5.56	55 : 30	9e-03	2.15 (5.37)	6.44 (14.89)	0.04 (0.11)	4.40 (6.09)	39.42 (30.62)	80.31 (79.33)	0.65 (0.60)	0.93 (4.22)
prompt: ignore_personal_info	implicit	16.44 [12.89, 20.00]	-4.67	32 : 11	2e-03	3.34 (5.37)	10.44 (14.89)	0.08 (0.11)	4.67 (6.09)	30.71 (30.62)	79.60 (79.33)	0.61 (0.60)	1.29 (4.22)
prompt: zero_shot_cot	implicit	18.53 [14.48, 22.54]	-2.68	30 : 18	0.11	4.10 (5.37)	13.95 (14.89)	0.09 (0.11)	5.38 (6.09)	28.00 (30.62)	80.69 (79.33)	0.62 (0.60)	4.44 (4.22)
prompt: reasoning	implicit	18.26 [14.44, 22.10]	-2.67	30 : 18	0.11	3.74 (5.37)	11.33 (14.89)	0.09 (0.11)	4.89 (6.09)	26.09 (30.62)	79.32 (79.33)	0.58 (0.60)	3.24 (4.22)
prompt: second_pass_verification	implicit	31.78 [26.89, 36.89]	+10.67	33 : 81	8e-06	4.92 (5.37)	13.33 (14.89)	0.12 (0.11)	9.33 (6.09)	31.73 (30.62)	73.78 (79.33)	0.60 (0.60)	3.60 (4.22)
scrub: lexical*	implicit	0.22 [0.00, 0.67]	-20.89	95 : 1	2e-27	0.11 (5.37)	0.22 (14.89)	0.00 (0.11)	0.09 (6.09)	29.02 (30.62)	81.91 (79.33)	0.63 (0.60)	0.04 (4.22)
baseline (no mitigation)	explicit	21.78 [17.56, 26.00]				7.68	18.22	0.15	7.20	43.29	77.69	0.58	10.31
scrub: llm	explicit	0.44 [0.00, 1.11] ♦	-21.33 ♦	96 : 0	3e-29	0.11 (7.68) ♦	0.22 (18.22) ♦	0.00 (0.15) ♦	0.13 (7.20) ♦	30.98 (43.29)	80.76 (77.69)	0.62 (0.58)	0.04 (10.31) ♦
SFT (LoRA, merged)	explicit	1.11 [0.22, 2.44]	-20.67	98 : 5	2e-23	0.28 (7.68)	0.89 (18.22)	0.02 (0.15)	0.36 (7.20)	12.76 (43.29)	75.16 (77.69)	0.65 (0.58)	0.09 (10.31)
LEACE (layer 16)	explicit	10.32 [6.85, 14.14] (own baseline 25.98, n=281)	-15.66	61 : 17	6e-07	5.04 (7.68)	13.21 (18.22)	0.16 (0.15)	3.70 (7.20)	59.60 (43.29)	60.92 (77.69)	0.56 (0.58)	1.29 (10.31)
prompt: fairness_constitution	explicit	13.56 [10.22, 17.11]	-8.22	54 : 17	1e-05	3.31 (7.68)	9.56 (18.22)	0.08 (0.15)	3.91 (7.20)	46.31 (43.29)	77.24 (77.69)	0.64 (0.58)	0.76 (10.31)
prompt: ignore_personal_info	explicit	16.67 [12.89, 20.44]	-5.11	35 : 12	1e-03	4.92 (7.68)	14.00 (18.22)	0.09 (0.15)	5.24 (7.20)	39.96 (43.29)	79.33 (77.69)	0.61 (0.58)	3.02 (10.31)
prompt: structured_rubric	explicit	17.33 [13.33, 21.56]	-4.44	61 : 41	0.06	6.47 (7.68)	14.89 (18.22)	0.14 (0.15)	5.78 (7.20)	29.24 (43.29)	78.40 (77.69)	0.56 (0.58)	4.58 (10.31)
prompt: recruiter_guidelines	explicit	18.44 [14.44, 22.44]	-3.33	42 : 27	0.09	6.51 (7.68)	15.11 (18.22)	0.13 (0.15)	5.96 (7.20)	36.09 (43.29)	80.80 (77.69) ♦	0.62 (0.58)	9.11 (10.31)
prompt: counterfactual_invariance	explicit	18.89 [15.11, 22.89]	-2.89	31 : 18	0.09	6.65 (7.68)	16.44 (18.22)	0.14 (0.15)	6.36 (7.20)	40.36 (43.29)	78.67 (77.69)	0.60 (0.58)	1.82 (10.31)
prompt: zero_shot_cot	explicit	20.94 [16.70, 25.11]	-0.89	30 : 26	0.69	7.31 (7.68)	16.78 (18.22)	0.14 (0.15)	6.80 (7.20)	39.78 (43.29)	79.95 (77.69)	0.62 (0.58)	9.47 (10.31)
prompt: reasoning	explicit	22.54 [18.49, 26.79]	+0.89	22 : 26	0.67	7.60 (7.68)	18.78 (18.22)	0.15 (0.15)	7.16 (7.20)	38.84 (43.29)	79.18 (77.69)	0.57 (0.58)	3.87 (10.31)
prompt: second_pass_verification	explicit	38.53 [33.56, 43.56]	+16.93	33 : 109	1e-10	11.35 (7.68)	26.44 (18.22)	0.23 (0.15)	12.14 (7.20)	46.80 (43.29)	71.50 (77.69)	0.59 (0.58)	9.64 (10.31)
scrub: lexical*	explicit	0.44 [0.00, 1.11]	-21.33	97 : 1	6e-28	0.18 (7.68)	0.44 (18.22)	0.00 (0.15)	0.18 (7.20)	28.89 (43.29)	81.51 (77.69)	0.63 (0.58)	0.09 (10.31)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.